

1. Administrative & Study Identification

Full Study Title: A Randomized Phase 3, Double-Blind Study of Chemotherapy With or Without Pembrolizumab Followed by Maintenance With Olaparib or Placebo for the First-Line Treatment of BRCA non-mutated Advanced Epithelial Ovarian Cancer (EOC)
Short Title/Acronym: KEYLYNK-001 / ENGOT-ov43 / GOG-3036 **Protocol Number:** MK-7339-001 **NCT Number:** NCT03740165 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab (MK-3475), Olaparib (MK-7339), Chemotherapy (Carboplatin [Trade Name: Paraplatin, ATC: L01XA02], Paclitaxel), Bevacizumab **Phase of Development:** Phase III **Trial Status:** Active, Not Recruiting **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy of treatment with carboplatin/paclitaxel PLUS pembrolizumab and maintenance olaparib in women with BRCA non-mutated advanced epithelial ovarian cancer (EOC), fallopian tube cancer, or primary peritoneal cancer, specifically measuring Progression-Free Survival (PFS) in patients whose tumors express PD-L1 (CPS ≥ 10) and in the intention-to-treat population. **Secondary Objectives:** To evaluate Overall Survival (OS), Progression-Free Survival (PFS) assessed by blinded independent central review, PFS after next-line treatment, and safety.

2.2 Background & Rationale There is a significant unmet need for new treatment regimens for BRCA1/2-nonmutated advanced ovarian cancer. Pembrolizumab, a PD-1-blocking antibody, in combination with chemotherapy and followed by maintenance with the PARP inhibitor olaparib, aims to provide added clinical benefit over Standard of Care chemotherapy in the homologous recombination deficiency (HRD) negative subgroup. This is particularly relevant for the PD-L1 CPS ≥ 10 population, attempting to delay cancer progression and improve overall patient outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator **Blinding:** Double-blind **Randomization:** Randomized (into 3 treatment arms)

3.2 Planned Enrollment & Duration Target N: 1,367 patients
Number of Sites: Global (Multicenter, including locations across

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria** 1. Female patients ≥ 18 years of age with histologically confirmed, FIGO stage III or IV epithelial ovarian cancer, primary peritoneal cancer, or fallopian tube cancer. 2. BRCA non-mutated status. 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 4. Patients must have recently undergone primary debulking surgery, been eligible for primary debulking surgery, or been a potential candidate for interval debulking surgery.
- 4.2 Exclusion Criteria** 1. Mucinous, germ cell, or borderline ovarian tumors. 2. Known BRCA mutation. 3. Prior systemic anticancer therapies or prior treatment with a PARP inhibitor or PD-1/PD-L1 inhibitors.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** * **Arm 1:** Pembrolizumab 200 mg IV Q3W (up to 35 cycles) + Chemotherapy (up to 5 cycles of Carboplatin [Trade Name: Paraplatin, ATC Code: L01XA02] AUC 5 or 6 and Paclitaxel 175 mg/m²) followed by Maintenance Olaparib 300 mg PO BID starting Cycle 7. * **Arm 2:** Pembrolizumab 200 mg IV Q3W + Chemotherapy followed by Placebo for Olaparib PO BID. * **Arm 3:** Placebo for Pembrolizumab IV Q3W + Chemotherapy followed by Placebo for Olaparib PO BID. **Route of Administration:** Intravenous (IV) for Pembrolizumab and Chemotherapy; Oral (PO) for Olaparib. **Duration of Treatment:** Up to 35 cycles (approximately 2 years) for Pembrolizumab; up to 2 years for Olaparib maintenance (or until disease progression).
- 5.2 Concomitant & Prohibited Therapy Permitted:** Bevacizumab via IV infusion on Day 1 of each 3-week cycle at the Investigator's discretion. Docetaxel (75 mg/m² Q3W) may be used for participants who experience severe hypersensitivity to paclitaxel. **Prohibited:** Prior targeted immune checkpoint blockade and prior PARP inhibitors.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Core/Excisional Biopsy for BRCA/PD-L1 status testing, Baseline Imaging
Baseline	Day 1	Randomization, Lead-in Cycle 1 (Chemotherapy)
Treatment / Interim	Every 3 Weeks (Q3W)	IV Infusions (Pembrolizumab/Placebo + Chemo), Safety Labs, Efficacy Assessment (RECIST 1.1)
Maintenance		

Start | Cycle 7 | Initiation of Olaparib or Placebo oral tablets
| | **End of Study** | Up to 57 Months | Final Imaging, Disease
Progression Assessment, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) in patients whose tumors expressed PD-L1 (Combined Positive Score [CPS] ≥ 10) and PFS in the intention-to-treat (ITT) population. **Measurement Method:** Assessed by the Investigator per RECIST 1.1 criteria.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) in patients with PD-L1 CPS ≥ 10 and the ITT population. 2. PFS per RECIST 1.1 assessed by Blinded Independent Central Review (BICR). 3. PFS after next-line treatment and safety.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and patient diaries; specifically monitoring immune-related AEs and toxicities associated with chemotherapy and PARP inhibitors. **Serious Adverse Events (SAEs):** Required reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee established to routinely assess safety and efficacy parameters throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population; Subgroup analysis for PD-L1 CPS ≥ 10 population; Safety Set. **Sample Size Justification:** Target $N=1,367$, powered to detect a statistically significant and clinically meaningful improvement in PFS and OS. **Handling of Missing Data:** Censoring applied for PFS per standard RECIST 1.1 guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:** Independent central review of imaging data and strict electronic Case Report Form (eCRF) tracking.

11. Study Identifiers & Governance

Primary ID: MK-7339-001 **Registry Number:** NCT03740165 **Posting ID:** 357438156295576930 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** KEYLYNK-001 / ENGOT-ov43 / GOG-3036 **Official Regulatory Title:** A Randomized Phase 3, Double-Blind Study of Chemotherapy With or Without Pembrolizumab Followed by Maintenance With Olaparib or Placebo for the First-Line Treatment of BRCA non-mutated Advanced Epithelial Ovarian Cancer (EOC)

12. Clinical Framework (Ovarian Cancer Specific)

Primary Condition: Advanced Epithelial Ovarian Cancer, Fallopian Tube Cancer, or Primary Peritoneal Cancer (Disease Areas: Ovarian Cancer, Fallopian Tube Cancer, Peritoneal Neoplasms) **Preferred UMLS Name:** Advanced Ovarian Cancer **Disease Terminology / Codes:** Neoplasms (Semantic Type: Neoplastic Process | MeSH: D009369 | HPO: HP:0002664 | SNOMED: 108369006) **Diagnosis Threshold:** FIGO Stage III-IV, BRCA non-mutated **Medical Methodology:** Immune Checkpoint Inhibition (PD-1) + Chemotherapy + PARP Inhibitor Maintenance **Optimization Target:** Delaying cancer progression (Progression-Free Survival) and extending Overall Survival

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for first-line systemic treatment post-debulking surgery in advanced ovarian cancer. **Education Protocol (HCP):** Protocol training on RECIST 1.1 evaluation, identification of immune-related AEs, and hypersensitivity management (e.g., transition to docetaxel). **Education Protocol (Patient):** Treatment adherence for oral maintenance phase (olaparib), side effect self-reporting. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all progression imaging events.

14. Observation & Follow-Up Schedule

Total Duration: Up to 57 months **Onsite Visit Cadence:** Every 3 weeks (Q3W) corresponding to IV infusion cycles. **Remote Monitoring Frequency:** Continuous safety monitoring as needed between cycles. **Checklist Review Interval:** Regular imaging and survival tracking updates post-treatment discontinuation.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |